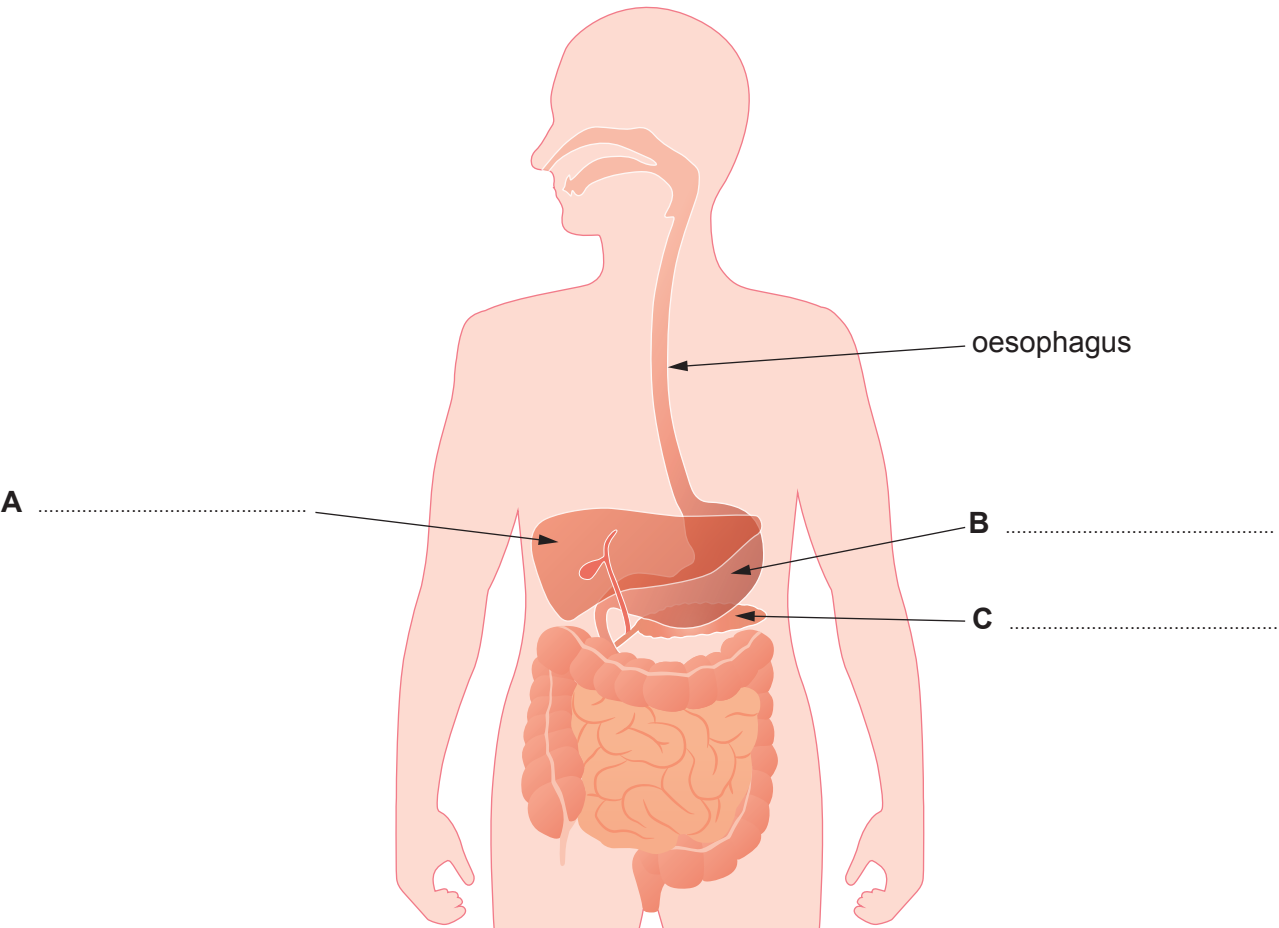


5. Image 5.1 shows part of the digestive system.

Image 5.1



(a) Label parts A to C on Image 5.1. [2]

(b) Explain why carbohydrase cannot digest fat. [1]

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- (c) Scientists investigated absorption of three different sugars in a sample of healthy small intestine in the laboratory. They analysed the absorption of the sugars in the presence or absence of cyanide. Cyanide is a chemical that prevents respiration taking place in cells.

The results are shown in **Table 5.2**.

Table 5.2

Sugar	Absorption without cyanide (a.u.)	Absorption with cyanide (a.u.)
glucose	100	33
xylose	30	30
arabinose	29	29

Use the information in **Table 5.2** to answer the following questions.

- (i) State which sugar is absorbed by active transport. Explain your answer. [4]

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- (ii) One of the scientists stated that all three of the sugars in the investigation could be absorbed by diffusion. Explain how the evidence in **Table 5.2** supports this conclusion. [2]

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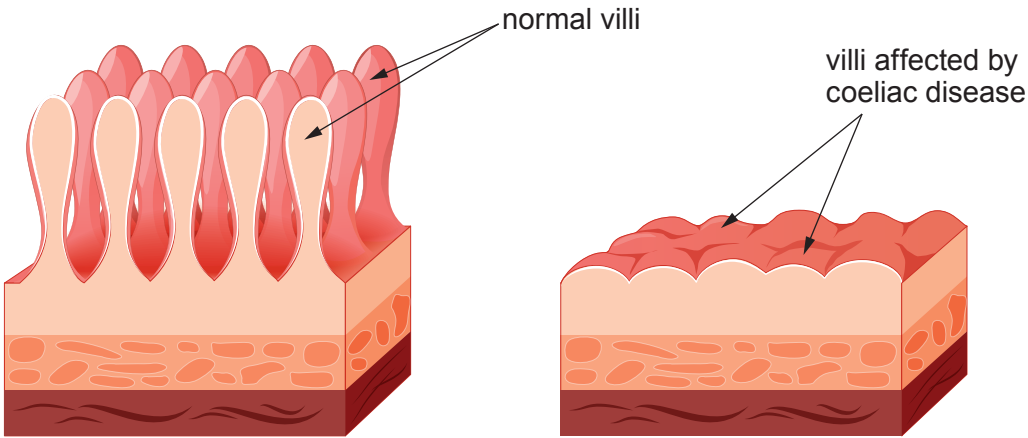
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(d) **Image 5.3** shows the structure of the lining of the small intestine of a healthy individual and an individual with coeliac disease. Coeliac disease causes a flattening of the villi lining the small intestine.

Image 5.3



Use **Image 5.3** to explain why individuals who have coeliac disease are unable to fully absorb their digested food. [1]

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